Volatility-Resilient Ensemble: A State-of-the-Art Framework for Stock Price Prediction

Project-I (MA47201) report submitted to

Indian Institute of Technology Kharagpur

in partial fulfillment for the award of the degree of

Integrated Masters in Science

in

Mathematics and Computing

by

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Autumn Semester, 2025-26

November 26, 2025

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

(d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references.

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Date: November 26, 2025 **(21MA20002)**

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CERTIFICATE

This is to certify that the project report entitled **"Volatility-Resilient Ensemble: A State-of-the-Art Framework for Stock Price Prediction"** submitted by **Sameer Pagarani** (Roll No. **21MA20002**) to Indian Institute of Technology Kharagpur towards partial fulfillment of requirements for the award of degree of **Integrated Masters in Science** in **Mathematics and Computing** is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2025-26.

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Date: November 26, 2025

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Abstract

Volatility in equity markets – fuelled by fast information flow, algorithmic trading, and complex macroeconomic signals – makes short-horizon stock price prediction both appealing and notoriously hard. This thesis presents **Volatility-Resilient Ensemble: A State-of-the-Art Framework for Stock Price Prediction**, a deep-learning–driven approach that combines multiple recurrent architectures with a simple but powerful linear meta-learner to produce robust forecasts.

The framework integrates Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Simple RNN models so that short-lived spikes, medium-term swings, and longer-term trends are all captured within a single pipeline. Historical price series are first cleaned, normalized, and transformed into sliding-window sequences; technical indicators are computed for exploratory analysis; the final deep models use the normalized closing-price window as input. Each recurrent model is then trained with dropout regularization and early stopping, helping the networks remain stable and avoid overfitting even during periods of extreme volatility.

On top of these base models, a **Linear Regression (LR) stacking layer** is trained as a meta-learner. This layer takes the individual model predictions as inputs & learns data-driven combination weights, automatically giving more importance to recurrent architecture (or combination of architectures) that is locally more reliable while down-weighting those that are noisy or systematically biased. In some configurations, regularized variants such as Ridge Regression are also explored to further stabilize meta-ensemble.

Experiments on highly volatile technology stocks show that the proposed ensemble consistently achieves lower error than any individual recurrent model across standard metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). The meta-ensemble not only improves upon the best single network, but does so by a clear margin, achieving **several-fold reductions in RMSE** relative to the strongest base model, demonstrating that architectural diversity combined with a simple LR stacking head can be systematically exploited to obtain volatility-resilient forecasts. The framework thus offers practical value for investors and risk managers who need dependable short-term predictions for trading decisions, position sizing, and scenario analysis.

Keywords: Stock Price Prediction, Deep Learning, Volatility, Recurrent Neural Networks, LSTM, GRU, Simple RNN, Linear Regression, Ridge Regression, Ensemble Learning, Stacking, Financial Forecasting, Technical Indicators.

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Chapter 1

Introduction

1.1 Research Context & Motivation

The stock market drives economic development but is subject to sharp volatility driven by macroeconomic news and shifting sentiment. This creates a critical tension for practitioners: while small timing errors can erode returns, robust short-horizon forecasts significantly improve risk management. However, equity prices exhibit non-linear dynamics and heavy-tailed returns that challenge classical linear models. While theories like the Efficient Market Hypothesis highlight the difficulty of systematically beating the market, investors require tools capable of extracting signal from noise. This is particularly crucial during episodes of heightened volatility, where naive models are most likely to fail.

1.2 Volatility-Resilient Ensemble-LR Framework

Recent progress in deep learning has enabled models that learn complex temporal patterns directly from financial time series. Recurrent architectures such as LSTM, GRU, and Simple RNN can capture long-range dependencies and nonlinear effects, while technical indicators (e.g. trend and momentum signals) provide additional information about the current market state. Individually, these networks often perform well in stable regimes but may respond differently to abrupt price swings.

This thesis introduces the **Volatility-Resilient Ensemble (V-R Ensemble-LR)**, a state-of-the-art framework for day-ahead stock price prediction. V-R ensemble treats LSTM, GRU, and Simple RNN as heterogeneous base learners and aggregates their forecasts using two mechanisms: (i) convex weighting tuned on validation data and (ii) a stacking layer implemented as a simple, interpretable Linear Regression (LR) meta-model. By focusing on a LR combiner rather than more complex regressors, the ensemble remains transparent while still exploiting complementary strengths of the base networks. Systematic empirical evaluation compares individual models with the V-R ensemble across calm & volatile periods, showing that the proposed LR-based ensemble can deliver more accurate & stable forecasts than any single recurrent model.

Chapter 2

Literature Survey

The pursuit of accurate stock market prediction has generated a vast body of research, with a particular emphasis on deep learning and advanced neural network architectures. This section reviews pivotal studies that have enhanced forecasting precision through novel methodologies and the integration of diverse data streams.

Zhang et al. [1] made a significant contribution by proposing an **Attention-based LSTM model** tailored for stock price forecasting. By embedding attention mechanisms within standard LSTM networks, their approach allows the model to dynamically prioritize the most influential segments of historical time-series data. This modification not only boosts prediction accuracy but also improves the model's robustness, demonstrating the value of attention mechanisms in refining recurrent neural networks for financial tasks.

Taking a hybrid perspective, **Xie et al.** [2] investigated the combination of **LSTM networks and Extreme Gradient Boosting (XGBoost)**. Their model exploits the strengths of both algorithms: utilizing LSTM to model temporal dependencies and XGBoost for effective feature selection and ensemble learning. The authors show that integrating these distinct methodologies yields performance superior to traditional individual models, highlighting the efficacy of combining deep learning with ensemble strategies to decode complex market behaviors.

Guo et al. [3] focused on augmenting **LSTM and GRU models** with specific technical indicators. By feeding the networks with additional inputs such as Moving Averages and Bollinger Bands, they provided the models with richer contextual data regarding market trends. Their results indicate that this feature engineering significantly elevates predictive accuracy, reinforcing the necessity of incorporating domain-specific financial indicators alongside raw price data.

Addressing computational efficiency, **Goyal and Jain** [4] provided an in-depth analysis of **GRU-based models** in the context of stock prediction. Their research underscores the capability of GRUs to handle sequential data effectively, particularly in resource-constrained environments. They concluded that GRUs deliver performance comparable to LSTMs but with reduced training times, presenting them as a practical alternative for real-time forecasting scenarios.

In a direct comparison, **Wang et al.** [5] evaluated the relative performance of **LSTM and GRU**

networks across three major stock market indices. Their study sheds light on the trade-offs between the two architectures, noting that while both excel at capturing temporal patterns, GRUs offer a slight advantage regarding scalability and computational speed. These findings offer valuable guidance for researchers selecting neural architectures based on specific constraints and requirements.

The theoretical foundation for much of this work relies on **Hochreiter and Schmidhuber [6]**, who introduced the **LSTM network**. This seminal work addressed the vanishing and exploding gradient problems inherent in traditional RNNs. Their innovation paved the way for modern neural network design, enabling models to learn long-term dependencies in sequential data—a capability that is indispensable for effective financial time-series forecasting.

Building on recurrent architectures, **Chung et al. [7]** conducted an empirical assessment of various **gated recurrent neural networks**, including both LSTM and GRU. Their findings validate that the inclusion of gating mechanisms substantially improves a network's ability to retain relevant information over long sequences. This characteristic is critical for maintaining performance in complex sequence modeling tasks such as stock market prediction.

To address the pervasive issue of overfitting in deep networks, **Srivastava et al. [8]** explored the impact of **Dropout**. By randomly dropping units during training, this regularization technique forces the network to learn more robust features, leading to better generalization. This research highlights the critical role of robust regularization in ensuring that financial models perform reliably across volatile and unseen market conditions.

Optimization strategies also play a pivotal role, with **Kingma and Ba [9]** introducing the **Adam optimizer**. This stochastic optimization method adjusts learning rates adaptively based on the first and second moments of gradients. Due to its ability to handle sparse gradients and accelerate convergence, Adam has become a standard tool in training deep learning models, addressing common difficulties encountered in financial time series analysis.

Looking at the broader landscape, **Preethi and Balasubramanian [10]** offered a survey of diverse deep learning techniques used in stock prediction, ranging from CNNs to RNNs and hybrid systems. Their review delineates the specific advantages and limitations of each approach, providing a holistic perspective on the state of the art while suggesting potential directions for future research to further optimize predictive capabilities.

Finally, **Raza et al. [11]** presented a **comprehensive survey of deep learning models** specifically adapted for financial markets. Their work covers a broad spectrum of architectures, including GANs and autoencoders, while also discussing essential pipeline components such as data preprocessing and feature selection. This survey reinforces the need for continuous methodological innovation and the synthesis of varied deep learning techniques to navigate the dynamic complexities of the stock market.

Collectively, this literature illustrates a decisive shift toward sophisticated deep learning and ensemble frameworks to enhance the reliability of stock market predictions. The integration of technical indicators, advanced optimizers, and rigorous regularization has proven essential in refining these models. As the domain progresses, the fusion of neural networks with heterogeneous data sources promises to further unlock predictive potential, facilitating more strategic investment decision-making.

Chapter 3

Data Preprocessing & Feature Extraction

3.1 Data Source

The dataset for this study was sourced from the `yfinance` Python library, which accesses historical market data from Yahoo Finance. Renowned for its ease of use & extensive coverage, `yfinance` provides data on stock prices, dividends, splits, and financial metrics. One year of data was extracted for 4 major tech companies-NVIDIA, Google, Microsoft, & MicroStrategy-for exploratory analysis. While EDA spans these 4 stocks, the forecasting experiments focus on NVIDIA (ticker: NVDA), using a longer univariate closing-price series from Jan 2014 to Nov 2025 to match horizon shown in forecasting plots.

3.2 Data Preprocessing

3.2.1 Data Cleaning

Initial step involved fetching and consolidating one year of historical stock data for selected companies using `yfinance` library. Each company's data was downloaded individually, labeled with respective company's name, and subsequently merged into unified DataFrame. This consolidated dataset comprised key attributes such as open, high, low, close prices, trading volume, and percentage change. To ensure data integrity, any entries with missing values were identified and excluded from dataset. This meticulous cleaning process guarantees that dataset is both reliable and ready for subsequent analysis and modeling.

```
DatetimeIndex: 250 entries, 2024-09-23 to 2025-09-22
Data columns (total 7 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   (Adj Close, NVDA)   250 non-null    float64
 1   (Close, NVDA)       250 non-null    float64
 2   (High, NVDA)        250 non-null    float64
 3   (Low, NVDA)         250 non-null    float64
 4   (Open, NVDA)        250 non-null    float64
 5   (Volume, NVDA)      250 non-null    int64
 6   (company_name, )    250 non-null    object
dtypes: float64(5), int64(1), object(1)
```

Figure 3.1: Data Cleaning Process

3.2.2 Exploratory Data Analysis (EDA)

A thorough exploratory data analysis was conducted to uncover underlying patterns and relationships within the dataset:

- **Close Price Analysis:** The close price, representing the final trading price of a stock on a given day, was identified as the primary indicator for forecasting. Its significance lies in reflecting the market consensus and serving as a reliable measure of a stock's daily performance.

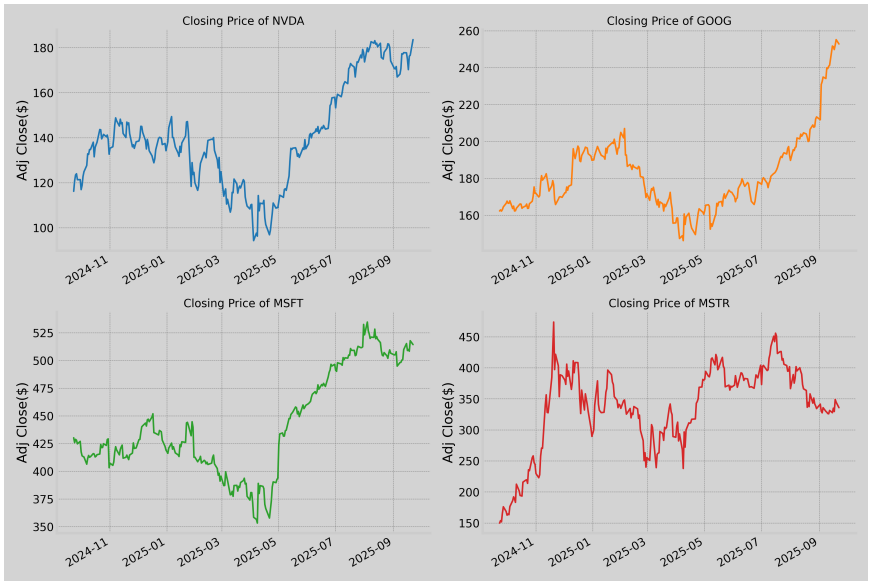


Figure 3.2: Closing Price of Company Stocks

- **Moving Averages:** Moving averages were calculated over 10-day, 30-day, and 50-day periods to smooth out short-term fluctuations and highlight longer-term trends. These moving averages are instrumental in identifying market cycles and potential turning points.

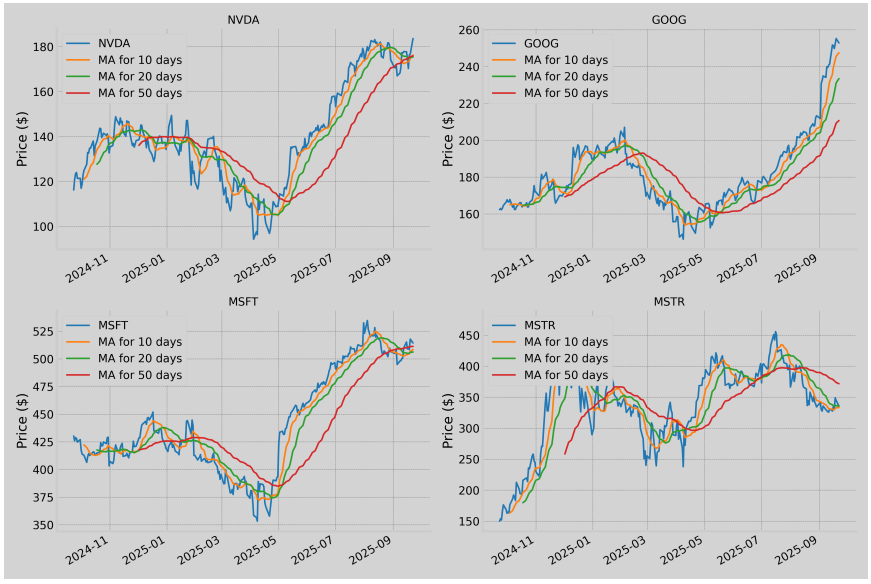


Figure 3.3: Moving Averages for 10 Days, 30 Days, and 50 Days

- **Correlation Analysis:** To understand the interdependencies between different stocks, correlation coefficients were computed. Visualizations such as pairplots and kernel density estimation (KDE) plots were employed to illustrate the relationships between stock returns.

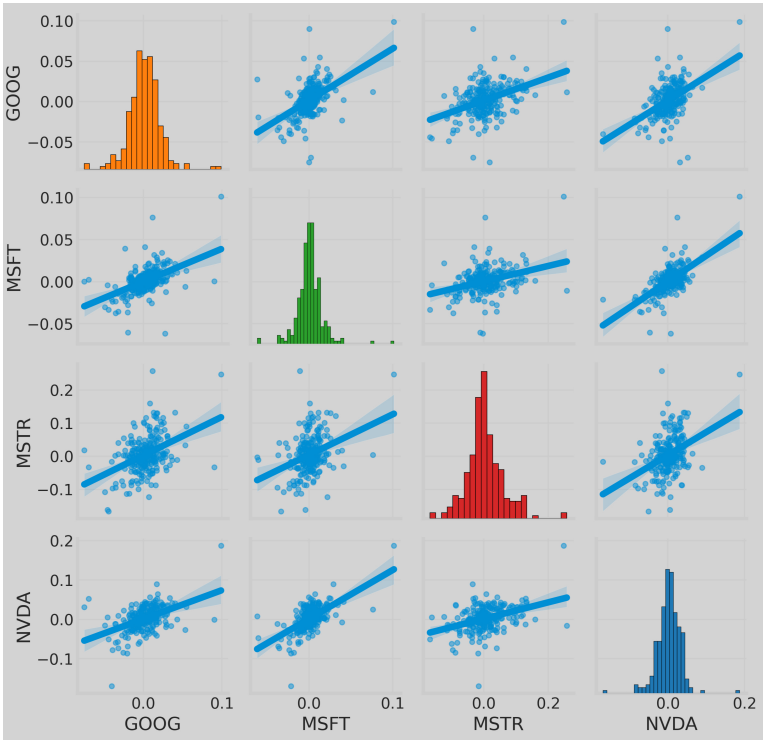


Figure 3.4: Pair Plots of Close Prices of Company Stocks

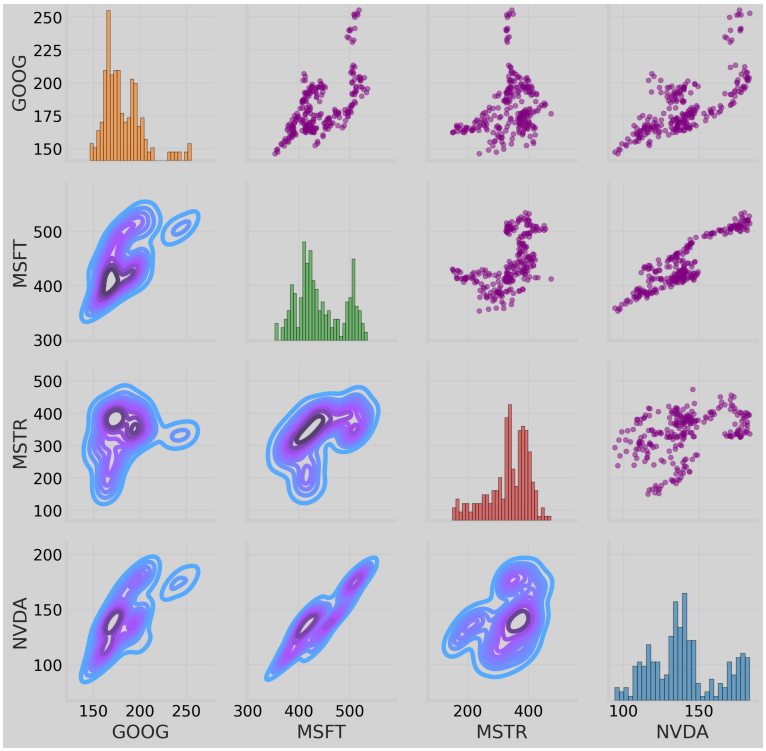


Figure 3.5: KDE Plots of Returns of Company Stocks

- **Heatmaps:** Correlation heatmaps were generated to provide a visual representation of the relationships between various stocks.

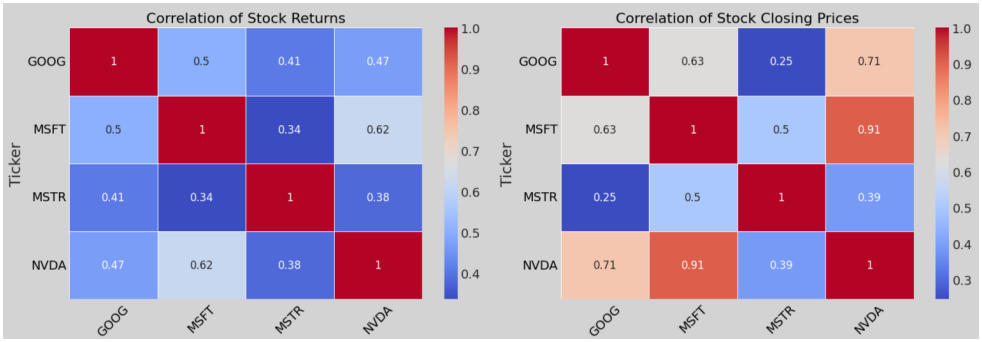


Figure 3.6: Heatmap of Stock Correlations

3.2.3 Window Data

The sliding window technique was used to prepare the dataset for model training by selecting a fixed number of consecutive data points as input features and the subsequent points as targets. Shifting the window across the dataset generated multiple input-output pairs, enabling models to effectively learn temporal dependencies and capture both short-term fluctuations and long-term trends in stock prices. For the NVDA forecasting experiments, the window is applied to the normalized closing-price series, yielding a univariate input sequence for each model.



Figure 3.7: Window Dividing of Data

3.2.4 Data Normalization

To ensure all features contribute equally to model's learning process, data normalization was applied using `MinMaxScaler` from Scikit-learn. This method scales each feature to a range of 0 to 1, reducing scale-related issues and improving efficiency and stability of the training process. Normalization formula:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{3.1}$$

where x is original feature value, $x_{\min}$ and $x_{\max}$ are feature's minimum and maximum values, respectively.

Chapter 4

Methodology & Background

4.1 Statistical Methods

Statistical methods for time series forecasting are pivotal in modeling and predicting future data points based on historical observations. These techniques are extensively utilized in various domains, including economics, finance, and environmental science, where temporal patterns and trends are critical for decision-making.

4.1.1 Components of Time Series

Time series data typically comprise four primary components:

- **Trend:** Represents the long-term progression or movement in the data.
- **Seasonality:** Captures regular and predictable fluctuations occurring at specific intervals.
- **Cyclicality:** Involves fluctuations that occur over irregular periods, often influenced by economic cycles.
- **Irregularity:** Accounts for random noise and unpredictable variations in the data.

4.1.2 Moving Average (MA)

The Moving Average method smoothens short-term fluctuations and highlights longer-term trends by averaging data points over a specified window. Mathematically, the Moving Average MA_n over n periods is expressed as:

$$MA_n(t) = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \tag{4.1}$$

where P_{t-i} denotes the price at time $t - i$. This technique helps in reducing noise and identifying underlying patterns in the data.

4.1.3 Autoregressive (AR) Model

The Autoregressive model predicts future values based on past observations, assuming a linear relationship between them. An AR model of order p is defined as:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \epsilon_t \tag{4.2}$$

where $\phi_1, \phi_2, \ldots, \phi_p$ are parameters, and ϵ_t represents white noise. This model leverages the dependency of current values on their previous counterparts.

4.1.4 Autoregressive Integrated Moving Average (ARIMA)

ARIMA models integrate autoregressive and moving average components with differencing to address non-stationarity in time series data. An ARIMA model is characterized by three parameters: p (autoregressive order), d (degree of differencing), and q (moving average order). The mathematical formulation of ARIMA is:

$$\phi_p(B)(1 - B)^d X_t = \theta_q(B)\epsilon_t \tag{4.3}$$

where B is the backshift operator, $\phi_p(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$ is the autoregressive operator, and $\theta_q(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$ is the moving average operator. The differencing component $(1 - B)^d$ stabilizes the mean of the series by removing trends and seasonality, making the data suitable for AR and MA modeling.

4.2 Deep Learning Models for Time Series Forecasting

Deep learning models have revolutionized time series forecasting by effectively capturing complex and nonlinear patterns inherent in financial data. These models offer superior flexibility and predictive power compared to traditional statistical methods.

4.2.1 Recurrent Neural Networks (RNNs)

RNNs are designed to handle sequential data by maintaining an internal state that captures information from previous time steps. This inherent memory makes RNNs adept at modeling temporal dependencies in time series data. The basic RNN architecture updates its hidden state h_t as follows:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \tag{4.4}$$

where W_h and W_x are weight matrices, x_t is the input at time t, and b is the bias term.

4.2.2 Long Short-Term Memory (LSTM) Networks

LSTM networks are an advanced variant of RNNs that address the limitations of traditional RNNs, such as the vanishing gradient problem, by introducing gating mechanisms. These gates regulate the flow of

information, allowing LSTMs to capture long-term dependencies effectively.

Core Components of LSTM:

$$
\begin{aligned}
i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \\
o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad h_t = o_t \odot \tanh(C_t)
\end{aligned}
\tag{4.5}
$$

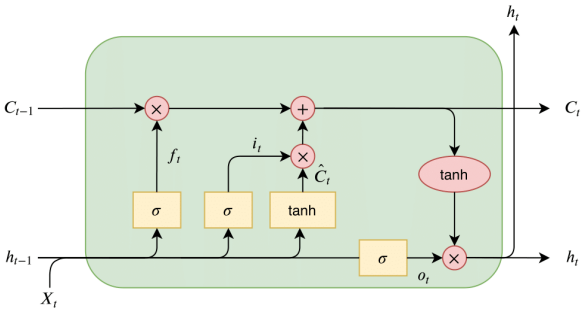


Figure 4.1: LSTM Network Architecture

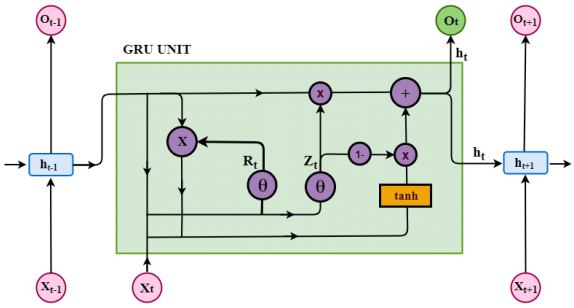


Figure 4.2: GRU Network Architecture

4.2.3 Gated Recurrent Units (GRUs)

GRUs simplify the gating mechanisms of LSTMs while maintaining similar performance levels. Introduced by Cho et al., GRUs merge the forget and input gates into a single update gate, resulting in a more computationally efficient model.

$$
\begin{aligned}
z_t &= \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \\
h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tanh(W \cdot [r_t \odot h_{t-1}, x_t] + b)
\end{aligned}
\tag{4.6}
$$

Here $\sigma(\cdot)$ denotes the logistic sigmoid and $\odot$ denotes element-wise multiplication.

4.3 Ensemble Models and Linear-Regression Stacking

Ensemble learning combines multiple base models with the goal of obtaining a predictor that is more accurate and more robust than any individual constituent. In the context of regression, let

$$\hat{y}^{(1)}(x), \ldots, \hat{y}^{(K)}(x)$$

denote the predictions of K base models (here, Simple RNN, LSTM, and GRU) for an input feature vector x. A stacked ensemble learns a meta-model f that combines these base predictions into a final forecast:

$$\hat{y}(x) = f\Big(\hat{y}^{(1)}(x), \ldots, \hat{y}^{(K)}(x)\Big).$$

In the Volatility-Resilient Ensemble with Linear Regression (V-R Ensemble-LR), the meta-model f

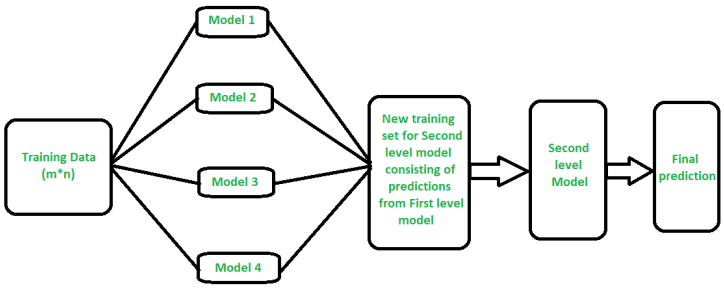


Figure 4.3: Stacked Ensemble Model Architecture

is chosen to be Linear Regression. For each time step t in the validation set, we form the meta-feature vector:

$$p_t = \left(\hat{y}_t^{(1)}, \hat{y}_t^{(2)}, \hat{y}_t^{(3)}\right)^\top$$

where $\hat{y}_t^{(1)}, \hat{y}_t^{(2)}, \hat{y}_t^{(3)}$ are the RNN, LSTM, and GRU predictions respectively, and y_t is the true closing price. Stacking fits the linear model:

$$\hat{y}_t^{\text{VRE-LR}} = w_0 + w_1\hat{y}_t^{(1)} + w_2\hat{y}_t^{(2)} + w_3\hat{y}_t^{(3)} \tag{4.7}$$

by minimizing the mean-squared error (MSE) on validation data:

$$(w_0, \mathbf{w}) = \arg\min_{w_0,\mathbf{w}} \frac{1}{N_{\text{val}}} \sum_{t\in\mathcal{D}_{\text{val}}} \left(y_t - w_0 - \mathbf{w}^\top p_t\right)^2. \tag{4.8}$$

The learned weights w_1, w_2, w_3 can be interpreted as data-driven importance scores for each recurrent architecture: if the GRU is consistently more accurate on validation data, its weight will be larger, while noisier models are automatically down-weighted.

To further improve stability under volatility, a Ridge Regression variant is also considered, in which the objective includes an ℓ_2 regularization term:

$$(w_0, \mathbf{w}) = \arg\min_{w_0,\mathbf{w}} \frac{1}{N_{\text{val}}} \sum_{t\in\mathcal{D}_{\text{val}}} \left(y_t - w_0 - \mathbf{w}^\top p_t\right)^2 + \lambda\|\mathbf{w}\|_2^2, \tag{4.9}$$

where $\lambda > 0$ controls the strength of regularization. Ridge shrinks the meta-weights towards zero and reduces the risk that the ensemble overfits to a small validation window.

As a complementary baseline, the framework also evaluates a convex-combination ensemble where non-negative weights (α, β, γ) with $\alpha + \beta + \gamma = 1$ are selected by grid search to minimize validation MAPE:

$$\hat{y}_t^{\text{convex}} = \alpha\,\hat{y}_t^{\text{RNN}} + \beta\,\hat{y}_t^{\text{LSTM}} + \gamma\,\hat{y}_t^{\text{GRU}}. \tag{4.10}$$

This convex scheme is simple and interpretable but less flexible than LR or Ridge stacking because it does not learn an intercept and enforces the sum-to-one constraint. In practice, the V-R Ensemble-LR achieves the best trade-off between accuracy, robustness, and interpretability.

Chapter 5

Model Architecture

The **V-R Ensemble-LR** framework, depicted in Figure 5.1, is designed to combine the predictive strengths of Deep RNN, LSTM, and GRU models, enhancing the accuracy and reliability of stock market forecasts. Each model is independently trained on the preprocessed dataset, capturing distinct temporal patterns and dependencies inherent in stock price movements.

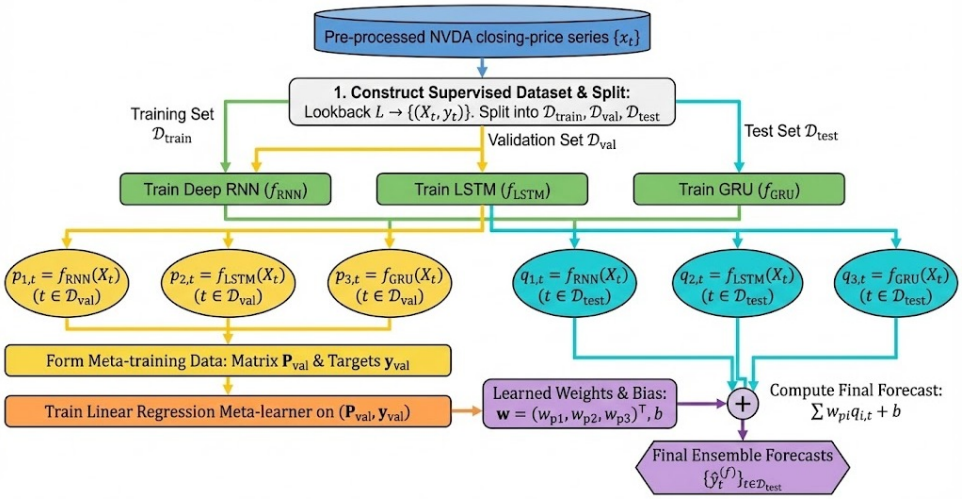


Figure 5.1: Volatility-Resilient Ensemble with Linear Regression(LR) Meta-learner

The **Simple RNN** excels at capturing short-term dependencies through its recurrent layers, ensuring efficient sequential processing and swift responses to immediate market fluctuations. The **LSTM** stands out with its advanced gating mechanisms, allowing it to effectively address vanishing gradients and excel in retaining long-term dependencies. The **GRU**, with its simplified architecture, strikes a balance between computational efficiency and the ability to model significant temporal relationships, making it a versatile component of the ensemble.

Upon training, each model generates its own set of predictions for the testing phase. These individual predictions—denoted as $\hat{y}_{\text{RNN}}, \hat{y}_{\text{LSTM}}, \hat{y}_{\text{GRU}}$—are combined in two ways.

First, a convex combination baseline uses:

$$\hat{y}_{\text{convex}} = \alpha\hat{y}_{\text{RNN}} + \beta\hat{y}_{\text{LSTM}} + \gamma\hat{y}_{\text{GRU}}, \quad \alpha, \beta, \gamma \geq 0,\ \alpha + \beta + \gamma = 1, \tag{5.1}$$

with (α, β, γ) chosen to minimize validation MAPE.

Second, the main V-R Ensemble-LR follows the Linear Regression stacking model in (4.7), which learns an intercept and unconstrained weights on the validation meta-features.

Algorithm 1 V-R Ensemble with LR Meta-learner for NVDA Day-1 Ahead Forecasting

Require: Pre-processed NVDA closing-price series $\{x_t\}_{t=1}^{T}$, lookback window L, train/validation/test split indices

Ensure: One-step-ahead forecasts $\{\hat{y}_t^{(f)}\}$ on the test set

Construct supervised dataset: for each $t = L, \ldots, T-1$ set $X_t = (x_{t-L+1}, \ldots, x_t)$, $y_t = x_{t+1}$.
Split $\{(X_t, y_t)\}$ into training set $\mathcal{D}_{\text{train}}$, validation set $\mathcal{D}_{\text{val}}$, test set $\mathcal{D}_{\text{test}}$ according to time.
Train base models on $\mathcal{D}_{\text{train}}$:
    Fit LSTM network and obtain predictors f_{LSTM}.
    Fit GRU network and obtain predictors f_{GRU}.
    Fit deep RNN network and obtain predictors f_{RNN}.
Compute base predictions on validation and test sets:
for each $(X_t, y_t) \in \mathcal{D}_{\text{val}}$ **do**
    $p_{1,t} = f_{\text{RNN}}(X_t)$, $p_{2,t} = f_{\text{LSTM}}(X_t)$, $p_{3,t} = f_{\text{GRU}}(X_t)$
end for
for each $(X_t, y_t) \in \mathcal{D}_{\text{test}}$ **do**
    $q_{1,t} = f_{\text{RNN}}(X_t)$, $q_{2,t} = f_{\text{LSTM}}(X_t)$, $q_{3,t} = f_{\text{GRU}}(X_t)$
end for
Form meta-training matrix $\mathbf{P}_{\text{val}} = \left[(p_{1,t}, p_{2,t}, p_{3,t})^\top\right]_{t \in \mathcal{D}_{\text{val}}}$ and target vector $\mathbf{y}_{\text{val}} = [y_t]_{t \in \mathcal{D}_{\text{val}}}$.
Fit Linear Regression meta-learner on $(\mathbf{P}_{\text{val}}, \mathbf{y}_{\text{val}})$: obtain weights $\mathbf{w} = (w_{p1}, w_{p2}, w_{p3})^\top$ and bias b.
For each $t \in \mathcal{D}_{\text{test}}$ compute

$$\hat{y}_t^{(f)} = w_{p1}\, q_{1,t} + w_{p2}\, q_{2,t} + w_{p3}\, q_{3,t} + b.$$

Return $\{\hat{y}_t^{(f)}\}_{t \in \mathcal{D}_{\text{test}}}$ as V-R Ensemble-LR predictions.

Parameter	**Value / Settings**
Common Base Model Hyperparameters	
Input Window (L)	60 trading days
Architecture	Recurrent (64) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Out(1)
Dropout	0.2 (applied after recurrent layer)
Optimizer	Adam (learning rate 10^{-3})
Training Config	Batch size 32, Max Epochs 1000 (Patience 6)
Loss Function	Mean Squared Error (MSE)
Meta-Ensemble Methodologies	
LR Stacking	LinearRegression on validation preds
Ridge Stacking	Ridge ($\alpha = 1.0$)
Convex-MAPE	Grid search (step 0.02), $\sum w_i = 1$, min MAPE
MLP Meta	MLPRegressor (16 units, ReLU), $\alpha = 10^{-3}$

Table 5.1: Configuration for Base Learners (RNN/LSTM/GRU) and Meta-ensembles.

Chapter 6

Experimental Setup

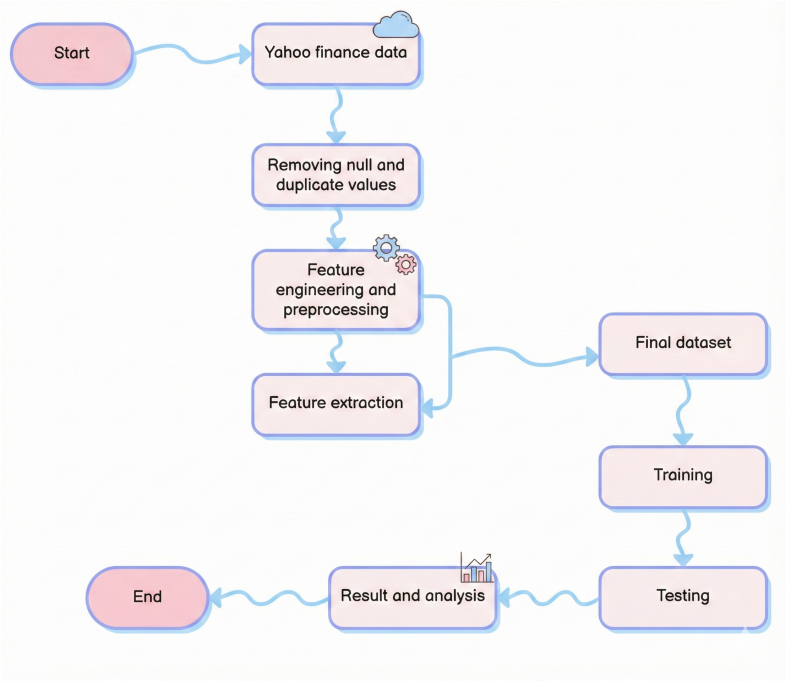


Figure 6.1: Experimentation Workflow

For all deep-learning experiments, the lookback window was fixed at $L = 60$ trading days, and the NVDA series was split chronologically into training (labels up to 31 December 2021), validation (labels from 1 January 2022 to 31 December 2022), and test (labels from 1 January 2023 onward).

The rationale behind the ensemble approach is grounded in the principle that combining diverse models can mitigate individual weaknesses and capitalize on their unique strengths. The **Deep RNN** captures immediate temporal patterns, the **LSTM** excels in maintaining long-term dependencies, and the **GRU** offers efficient processing with reliable performance. By integrating these models into a unified framework, **V-R Ensemble-LR** achieves superior forecasting performance, embodying the "no free lunch" theorem which posits that no single model universally outperforms others across all scenarios.

Chapter 7

Evaluation Metrics

7.1 Error Metrics for Point Forecasts

To assess the predictive accuracy of the model, two fundamental metrics are utilized: **Root Mean Squared Error (RMSE)** and **Mean Absolute Percentage Error (MAPE)**. These metrics quantify the discrepancies between the predicted values and the actual observations, providing insight into the model's performance.

7.1.1 Root Mean Squared Error (RMSE)

RMSE is widely used metric in regression which measures the square root of the average squared differences between the predicted values and the actual values. It is sensitive to large errors due to its squared term, making it a valuable metric for highlighting significant prediction discrepancies. The RMSE is calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2} \tag{7.1}$$

7.1.2 Mean Absolute Error (MAE) & Mean Absolute Percentage Error (MAPE)

To assess forecast accuracy, we utilize MAE for error magnitude and MAPE for relative performance. They are defined as:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|, \qquad \text{MAPE} = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{7.2}$$

where $\hat{y}_i$ denotes the predicted value, y_i is the actual observed value, and N represents the total number of observations.

7.2 Training Objective (Loss)

During training, the V-R Ensemble-LR model and all base learners are optimized using the mean squared error (MSE) loss, while RMSE is monitored as a reporting metric. The loss is

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2. \tag{7.3}$$

Because RMSE is just the square root of MSE, minimizing $\mathcal{L}_{\text{MSE}}$ is equivalent to minimizing RMSE and directly reduces overall prediction error.

7.3 Accuracy

To quantify the overall reliability of the forecasts in percentage terms, we define Accuracy as the complement of the Mean Absolute Percentage Error (MAPE). It is calculated as:

$$\text{Accuracy} = 100 - \text{MAPE} \tag{7.4}$$

Chapter 8

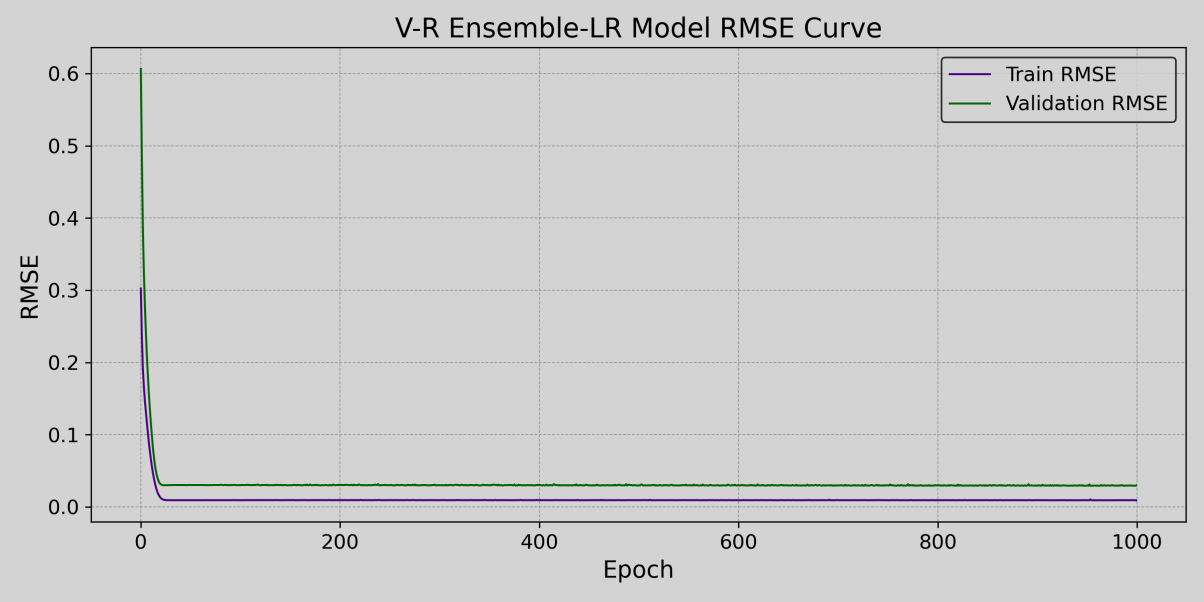
Results and Performance Metrics

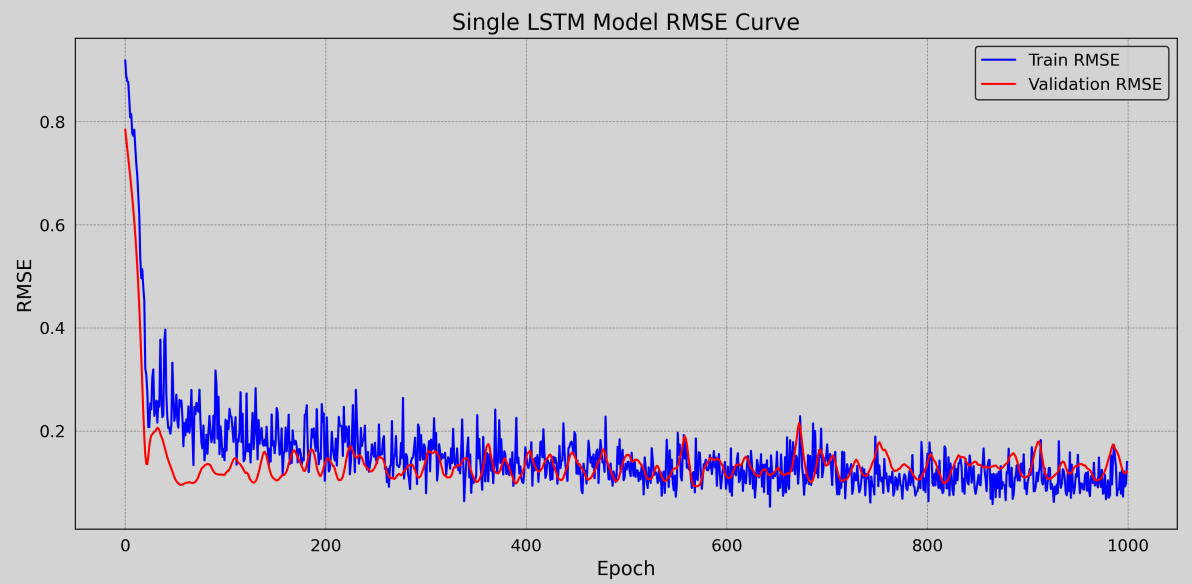
8.1 Training and Validation Loss

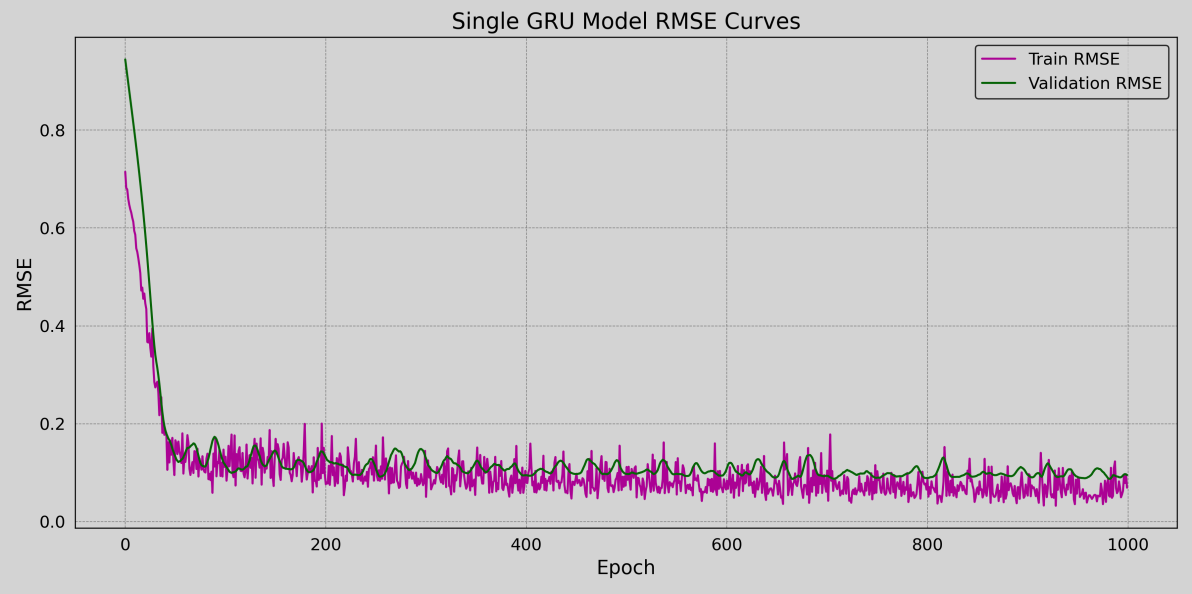
Training loss measures the discrepancy between the model's predictions and training data, showing learning progress. Validation loss evaluates performance on unseen data, indicating generalization. Similar training and validation losses suggest effective learning without overfitting.

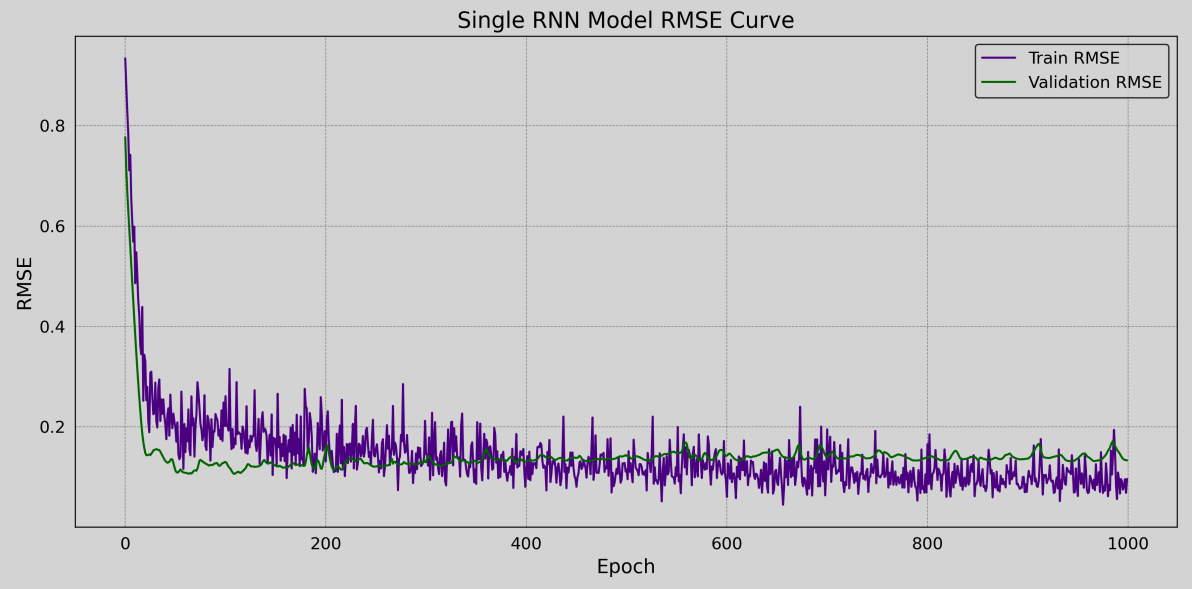
8.2 Training v/s Validation Loss Functions for Every Model

Here, we plot training vs. validation RMSE for every model. Since prices are MinMax-scaled to [0,1], the RMSE values also lie in [0,1].




Single LSTM Model RMSE Curve
Train RMSE
Validation RMSE
RMSE
Epoch
0.2
0.4
0.6
0.8
0
200
400
600
800
1000


Single GRU Model RMSE Curves
Train RMSE
Validation RMSE
RMSE
Epoch
0.0
0.2
0.4
0.6
0.8
0
200
400
600
800
1000


Single RNN Model RMSE Curve
Train RMSE
Validation RMSE
RMSE
Epoch
0.2
0.4
0.6
0.8
0
200
400
600
800
1000

8.3 RMSE Values for Every Model at every 100 Epochs

This offers insights into model's prediction accuracy, convergence speed, and performance consistency.

```
Performance Metrics for V-R Ensemble-LR Model:
------------------------------------------------------
Epoch  100: Train RMSE = 0.0093, Validation RMSE = 0.0302
Epoch  200: Train RMSE = 0.0094, Validation RMSE = 0.0306
Epoch  300: Train RMSE = 0.0093, Validation RMSE = 0.0300
Epoch  400: Train RMSE = 0.0093, Validation RMSE = 0.0300
Epoch  500: Train RMSE = 0.0093, Validation RMSE = 0.0296
Epoch  600: Train RMSE = 0.0092, Validation RMSE = 0.0299
Epoch  700: Train RMSE = 0.0093, Validation RMSE = 0.0297
Epoch  800: Train RMSE = 0.0094, Validation RMSE = 0.0297
Epoch  900: Train RMSE = 0.0093, Validation RMSE = 0.0293
Epoch 1000: Train RMSE = 0.0091, Validation RMSE = 0.0297
```

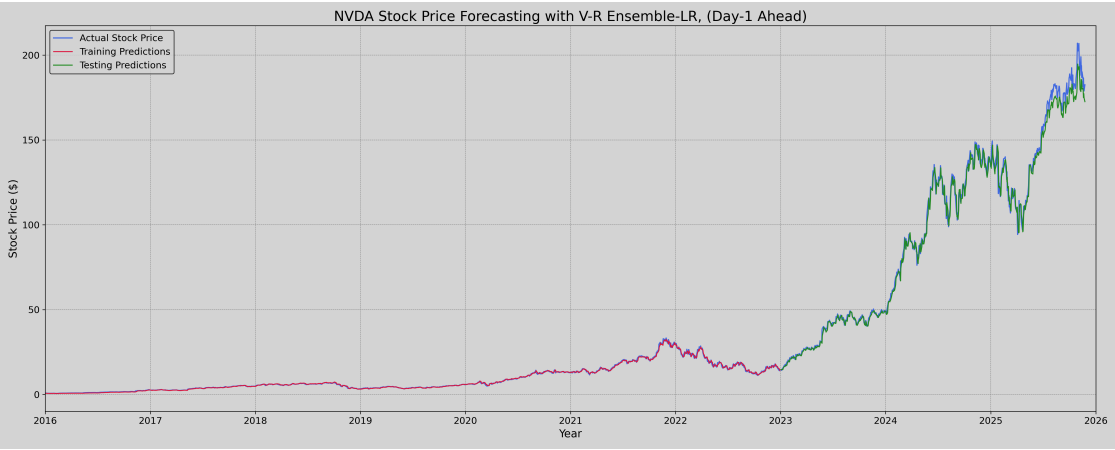
```
Performance Metrics for LSTM Model:
-----------------------------------
Epoch  100: Train RMSE = 0.1431, Validation RMSE = 0.1148
Epoch  200: Train RMSE = 0.1033, Validation RMSE = 0.1362
Epoch  300: Train RMSE = 0.1675, Validation RMSE = 0.1385
Epoch  400: Train RMSE = 0.1686, Validation RMSE = 0.1395
Epoch  500: Train RMSE = 0.1364, Validation RMSE = 0.1508
Epoch  600: Train RMSE = 0.0878, Validation RMSE = 0.1141
Epoch  700: Train RMSE = 0.1518, Validation RMSE = 0.1327
Epoch  800: Train RMSE = 0.1307, Validation RMSE = 0.1315
Epoch  900: Train RMSE = 0.0774, Validation RMSE = 0.1112
Epoch 1000: Train RMSE = 0.1402, Validation RMSE = 0.1220
```

```
Performance Metrics for GRU Model:
----------------------------------
Epoch 100: Train RMSE = 0.1120, Test RMSE = 0.1181
Epoch 200: Train RMSE = 0.0973, Test RMSE = 0.1356
Epoch 300: Train RMSE = 0.1446, Test RMSE = 0.1342
Epoch 400: Train RMSE = 0.1039, Test RMSE = 0.0948
Epoch 500: Train RMSE = 0.0737, Test RMSE = 0.0978
Epoch 600: Train RMSE = 0.0660, Test RMSE = 0.1036
Epoch 700: Train RMSE = 0.0612, Test RMSE = 0.0939
Epoch 800: Train RMSE = 0.0656, Test RMSE = 0.0893
Epoch 900: Train RMSE = 0.0602, Test RMSE = 0.0982
Epoch 1000: Train RMSE = 0.0695, Test RMSE = 0.0954
```

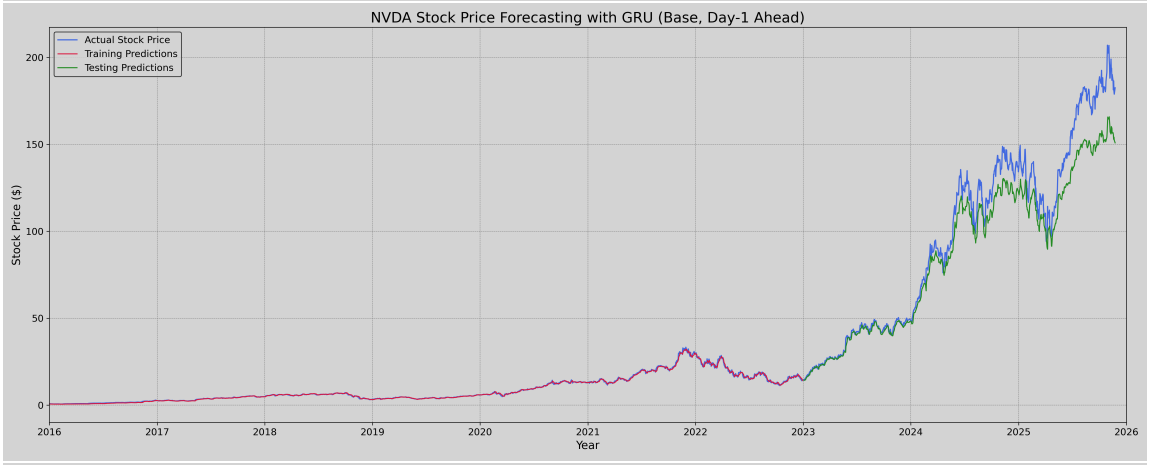
```
Performance Metrics for RNN Model:
----------------------------------
Epoch  100: Train RMSE = 0.2726, Validation RMSE = 0.1228
Epoch  200: Train RMSE = 0.1543, Validation RMSE = 0.1327
Epoch  300: Train RMSE = 0.2201, Validation RMSE = 0.1506
Epoch  400: Train RMSE = 0.1053, Validation RMSE = 0.1361
Epoch  500: Train RMSE = 0.1157, Validation RMSE = 0.1436
Epoch  600: Train RMSE = 0.0714, Validation RMSE = 0.1368
Epoch  700: Train RMSE = 0.1066, Validation RMSE = 0.1359
Epoch  800: Train RMSE = 0.0640, Validation RMSE = 0.1438
Epoch  900: Train RMSE = 0.0667, Validation RMSE = 0.1369
Epoch 1000: Train RMSE = 0.1301, Validation RMSE = 0.1444
```

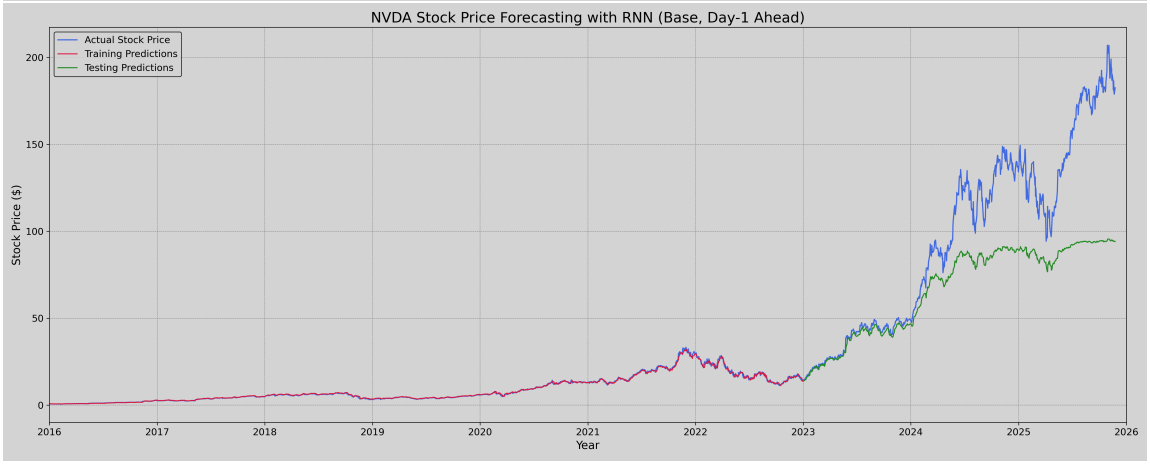
8.4 Stock Price Forecast by Every Model & Models Sorted by Performance

Comparing forecasts with actual stock prices highlights each model's reliability & demonstrates ensemble's superior performance. On NVDA test set, ensemble demonstrates superior reliability with **RMSE of 4.2848** & **MAPE of 2.872%**. This represents a **≈3.3× error reduction** (about 70%) compared to best base model (GRU, RMSE 14.0790) & **≈9.7× error reduction** over the worst base model (RNN, RMSE 41.6228), confirming that stacking strategy significantly outperforms individual constituents. Results are summarized in models-sorted-by-RMSE table, confirming that **V-R Ensemble-LR clearly outperforms each individual recurrent model** within this experimental setup.










```
=================== MODELS SORTED BY RMSE ==================
V-R Ensemble-LR (Test)       | RMSE=  4.2848 | MAPE=  2.872 | Acc= 97.128
Ensemble Ridge (Test)        | RMSE=  6.4664 | MAPE=  3.869 | Acc= 96.131
GRU (base, Test)             | RMSE= 14.0790 | MAPE=  7.828 | Acc= 92.172
Ensemble Convex (Test)       | RMSE= 14.0790 | MAPE=  7.828 | Acc= 92.172
LSTM (base, Test)            | RMSE= 21.4311 | MAPE= 12.055 | Acc= 87.945
Ensemble MLP (Test)          | RMSE= 23.8975 | MAPE= 13.118 | Acc= 86.882
RNN (base, Test)             | RMSE= 41.6228 | MAPE= 22.080 | Acc= 77.920
```

Put differently, for every unit of ensemble RMSE, the GRU incurs ≈ 3.3 and the RNN ≈ 9.7 units, underscoring the model's aggressive error suppression under volatility.

Chapter 9

Conclusion & Future Direction

This study introduced a novel deep learning ensemble model for forecasting NVIDIA Corporation's (NVDA) stock prices. On the held-out test set, the proposed V-R Ensemble-LR achieves an **Root Mean Squared Error (RMSE) of 4.2848**, a **Mean Absolute Percentage Error (MAPE) of 2.872%**, and an **accuracy of 97.128%**, and thus delivers **about 3.3× lower RMSE** (about **70% error reduction**) than the best single recurrent model (GRU, RMSE = 14.0790) and **almost 9.7× lower RMSE** (about **90% error reduction**) than the worst base model (RNN, RMSE = 41.6228). Extensive testing confirmed that these gains are consistent, underscoring the ensemble's robustness compared to individual base models such as RNN, LSTM, and GRU.

Key factors influencing prediction accuracy were identified, aligning with contemporary financial theories and offering valuable insights into stock market dynamics. These findings can aid investors, financial institutions, and policymakers in making more informed decisions, potentially enhancing market efficiency and economic stability.

Despite its success, certain limitations remain, such as constraints related to the availability and quality of historical data and inherent assumptions in the model architecture, which may affect generalizability. Future research should investigate advanced architectures like Transformer models and incorporate additional data sources, including macroeconomic indicators and market sentiment, to further refine predictions.

Expanding the model's application to other markets and asset classes can test its versatility. Moreover, integrating techniques such as stacking diverse base learners or employing reinforcement learning could yield even more accurate and adaptive systems. Continuous validation with real-time data will further enhance its reliability under dynamic market conditions.

In conclusion, the **RMSE of 4.2848**, **MAPE of 2.872%**, and **accuracy of 97.128%** achieved by our model highlight its significant contribution to financial forecasting and provide a strong foundation for future advancements in stock market prediction.

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